Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope, with an optional endurance limit, and the spectrum repetitions to failure. It excludes life below 10^4 cycles and a second S–N slope.

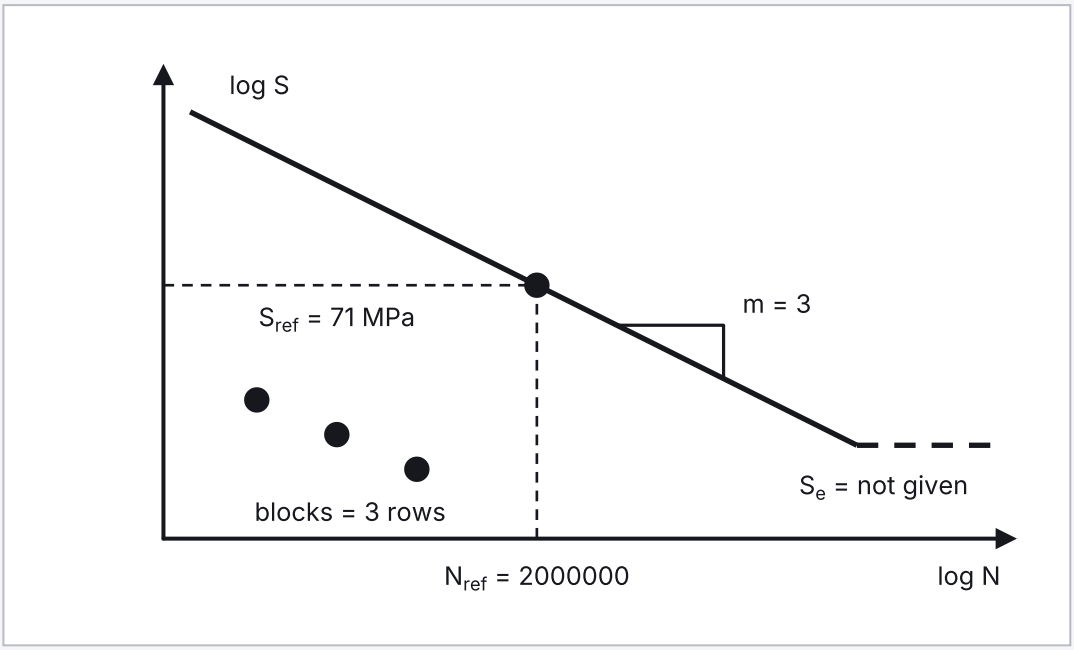


Figure 1 — A single-slope S–N line on log axes through its reference point, the endurance limit as a cut-off, and the stress-range blocks as listed (three drawn, representatively). Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Cycles to failure at each stress range, from the S–N curve (unbounded at or below the endurance limit S_e)	$N_i = N_{ref} \cdot (S_{ref}/S_i)^{\wedge}m$
Damage done by each block of cycles	$d_i = n_i/N_i$
Total damage, and how many repetitions of the spectrum the part survives	$D_{tot} = \Sigma\, d_i$; repetitions = $1/D_{tot}$
Acceptance by the Palmgren–Miner rule	$D_{tot} \le 1$

Basis: Fatigue resistance (S–N curves by detail category)[1](cl. 8); Cumulative fatigue damage[2](§6-15, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_{ref}	Reference stress range (detail category or material fatigue strength)	71	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{ref}	Cycles at the reference stress range	2000000		spec: SPEC-MECH-03 design criteria	10000 … 1000000000
m	S–N slope	3		spec: SPEC-MECH-03 design criteria	2 … 15
blocks	Stress-range blocks	3 row(s)		spec: LOAD-SCHEDULE-01	any

TABLE ROW	VALUES
blocks[1]	S = 100 MPa; n = 100000
blocks[2]	S = 60 MPa; n = 500000
blocks[3]	S = 40 MPa; n = 2000000

3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. The S–N curve has one slope over the whole range, with no second slope below a knee. (A simplified curve: enter a two-slope curve as two sheets or with a lower slope.)

4. Working

[4] $N_1 = N_{ref} \cdot (S_{ref} / S_1)^m$ [1](1993-1-9:2005 cl. 8)
$= 2000000 \times (71\ MPa / 100\ MPa)^3$
$= 715820$
Life at block 1

[5] $d_1 = n_1 / N_1$
$= 100000 / 715820$
$= 0.1397$
Damage of block 1

[6] $N_2 = N_{ref} \cdot (S_{ref} / S_2)^m$ [1](1993-1-9:2005 cl. 8)
$= 2000000 \times (71\ MPa / 60\ MPa)^3$
$= 3314000$
Life at block 2

[7] $d_2 = n_2 / N_2$
$= 500000 / 3314000$
$= 0.15088$
Damage of block 2

[8] $N_3 = N_{ref} \cdot (S_{ref} / S_3)^m$ [1](1993-1-9:2005 cl. 8)
$= 2000000 \times (71\ MPa / 40\ MPa)^3$
$= 11185000$
Life at block 3

[9] $d_3 = n_3 / N_3$
$= 2000000 / 11185000$
$= 0.17882$
Damage of block 3

[10] $D_{tot} = \Sigma\ d_i$ [2](§6-15)
$= 0.1397 + 0.15088 + 0.17882$
$= 0.46939$
Total damage (Palmgren–Miner)

[11] $D_{lim} = 1$ [2](§6-15)
= 1
Damage limit of the Palmgren–Miner rule
[12] $R_f = 1 / D_{tot}$
= 1 / 0.46939
= 2.1304
Spectrum repetitions to failure

5. Results

SYMBOL	VALUE	DESCRIPTION
D_{tot}	0.47	Total damage (governing)
R_f	2.1304	Repetitions to failure
N_1	715820	Life at block 1
d_1	0.1397	Damage of block 1
N_2	3314000	Life at block 2
d_2	0.15088	Damage of block 2
N_3	11185000	Life at block 3
d_3	0.17882	Damage of block 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Total damage against the damage limit	0.46939	1	0.47	[2](§6-15)

7. Applicability

None.

8. Verification

Template mech.fatigue-miner@1.1.0 Content hash b0537c083ea4cfc5 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-9: Fatigue. BS EN 1993-1-9. BSI.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.